

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 19

CLASSIFICATION OF WINDS

- **LOCAL WINDS**

GLOBAL PLANETARY DISTRIBUTION OF AIR PRESSURE

PRESSURE BELTS ON THE BASIS OF MODE OF GENESIS

THERMALLY INDUCED PRESSURE BELTS

1 This is due to the influence of temperature by which the pressure tends to rise from the equator towards the poles.

2 Belts produced because of this:
1. Equatorial low pressure
2. Polar high pressure

DYNAMICALLY INDUCED PRESSURE BELTS

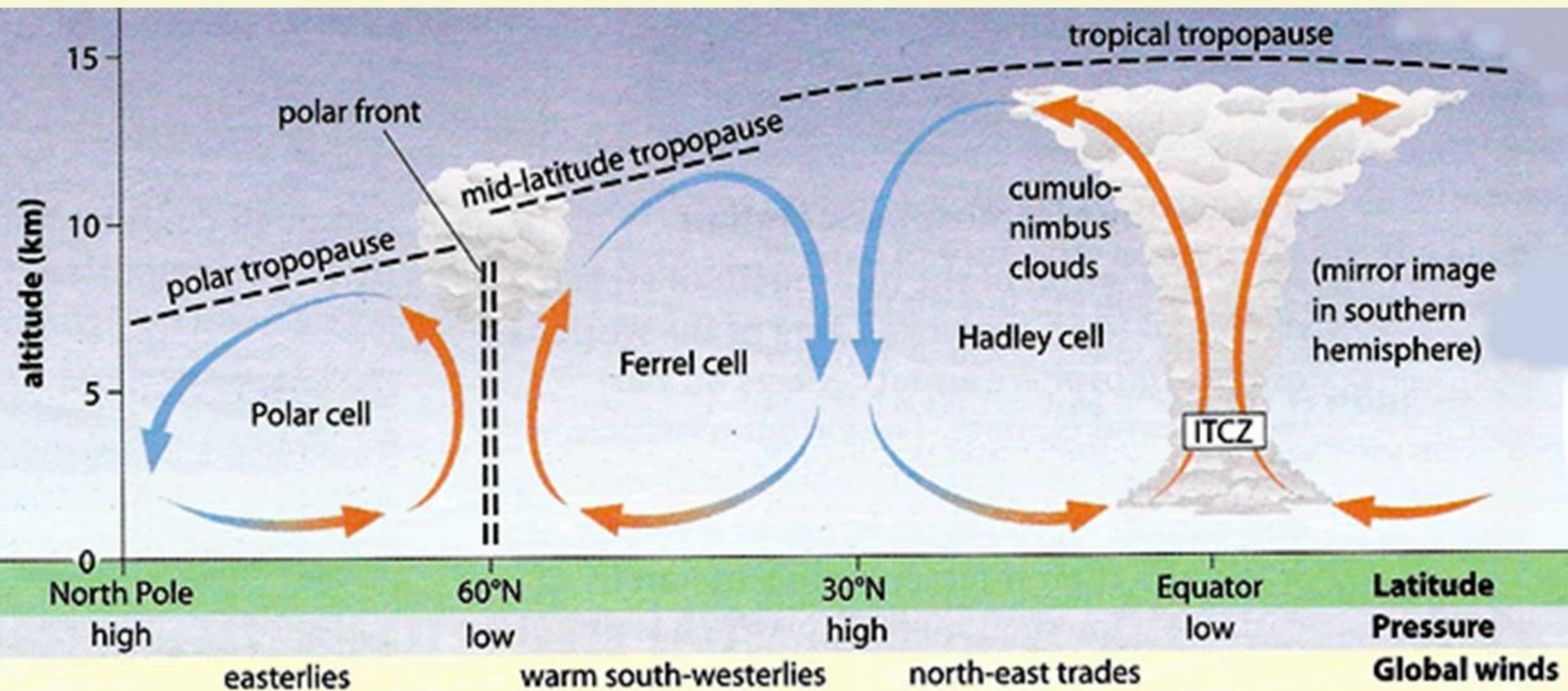
1 This is caused by the rotation of the earth by which air is pulled inwards in the polar region. This causes a rarification of the atmosphere in the sub-polar latitudes.

2 Belts produced because of this:
1. Sub-Tropical High Pressure Belt
2. Sub-Polar Low Pressure Belt

In all, there are seven pressure belts on the globe.



QUICK RELOOK AT MODULE 18



So to sum it up, we saw how the global pressure belts are distributed and how planetary winds originate because of the pressure belts.



ATMOSPHERIC PRESSURE

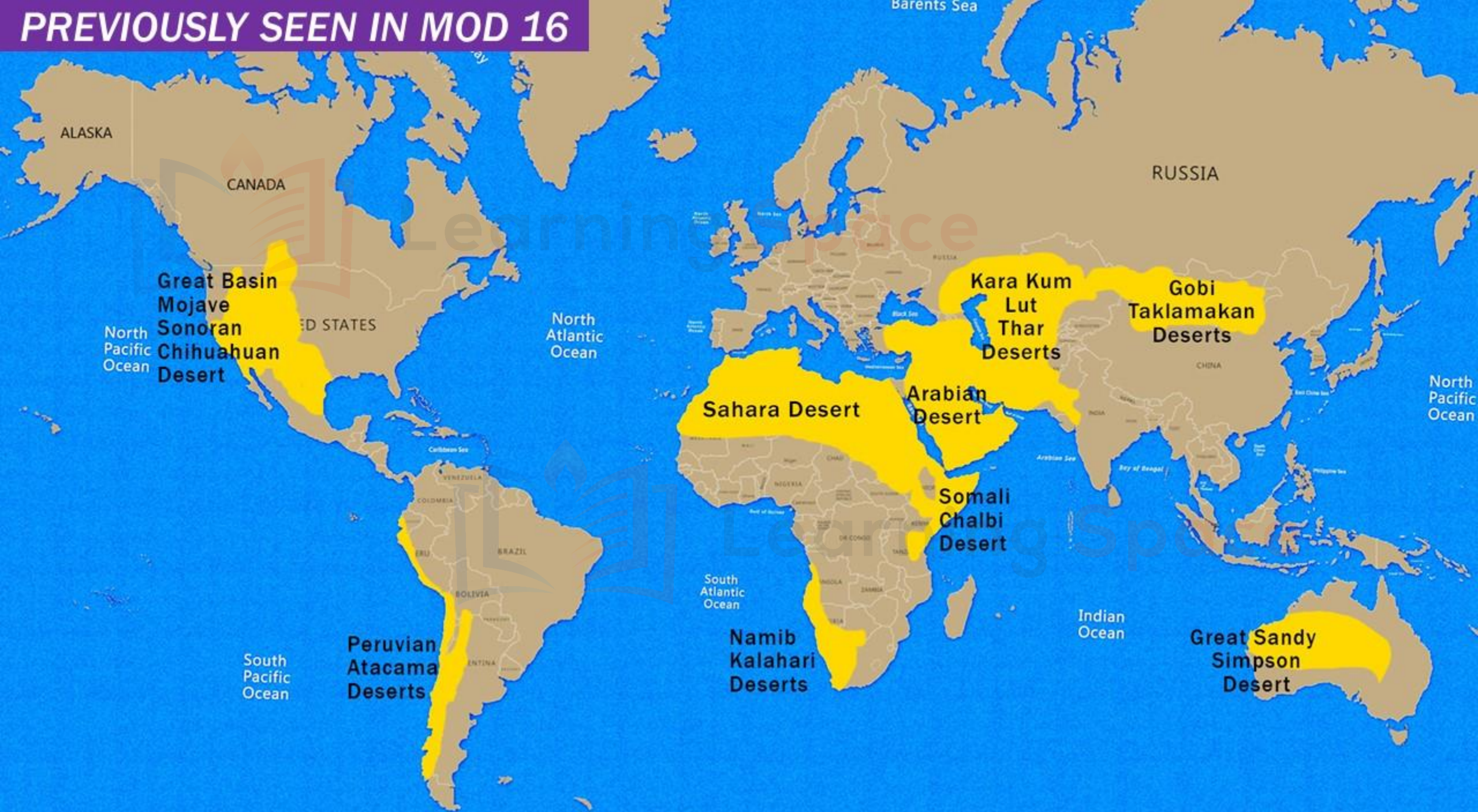
UPSC 2013 GS-1 QUESTION

Major hot deserts in northern hemisphere are located between 20-30 degree north and on the western side of the continents. Why? (10 Marks)

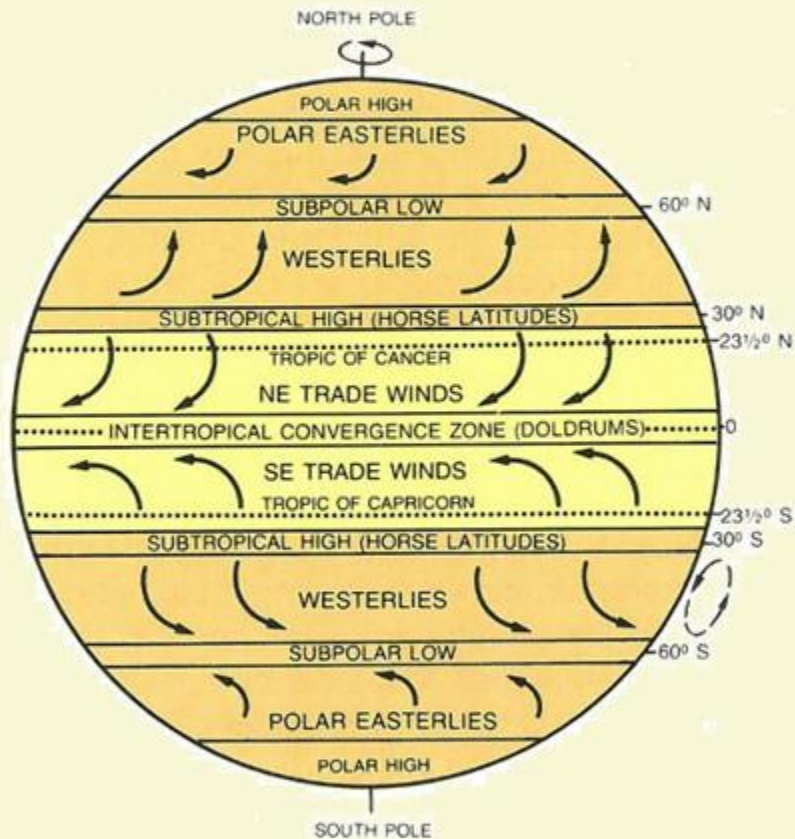
HINTS

- 1 Offshore trade winds in the region
- 2 Prevailing anticyclonic conditions- Areas between 20-30 degree latitudes on western margins of continents are the regions of descending air
- 3 Presence of cold ocean currents along the western coast of continents- this stabilizes the air and prevents cloud formation and rainfall.
- 4 Leeward sides of mountains/Parallel mountain ranges

PREVIOUSLY SEEN IN MOD 16



CLASSIFICATION OF WINDS

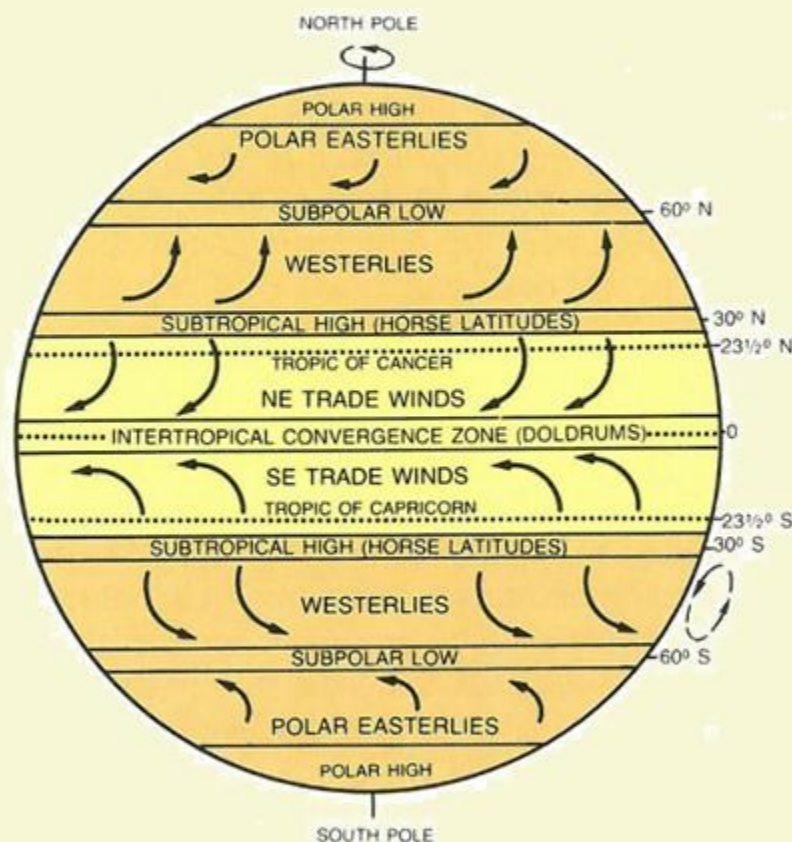


The winds are classified into two broad categories

- Permanent Winds or Prevailing Winds
- Variable winds

CLASSIFICATION OF WINDS

PERMANENT WINDS



1 The winds that **blow almost in the same direction throughout the year** are called Prevailing or Permanent winds. These are also called as **Planetary winds** because they involve large areas of the globe.

2 These include

1. Trade winds
2. Westerlies and
3. Polar Easterlies

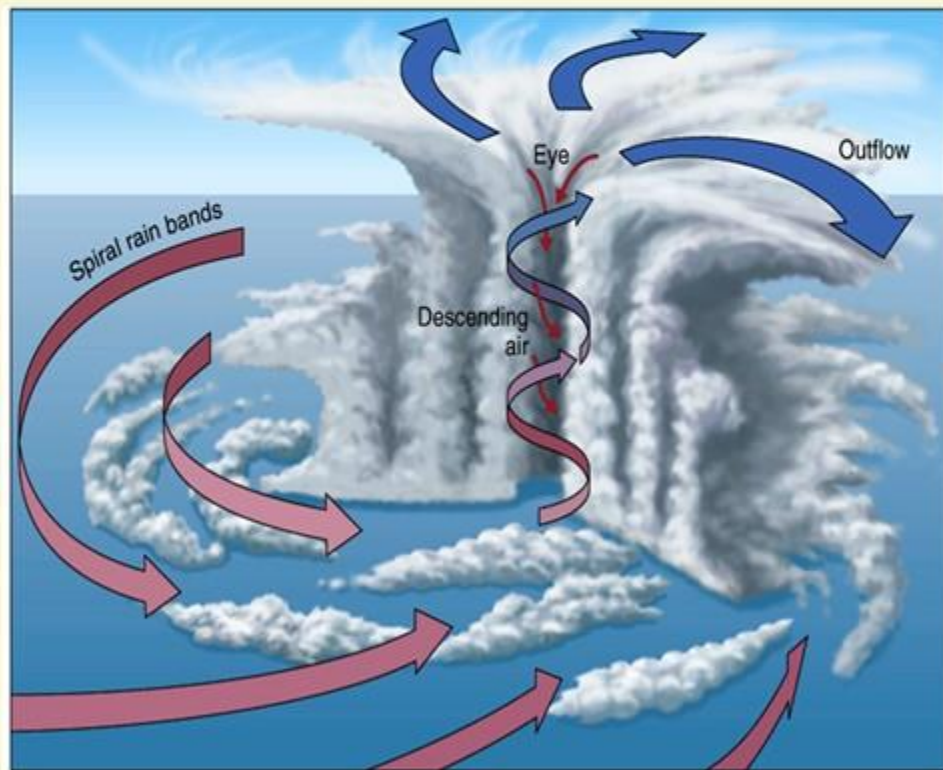
CLASSIFICATION OF WINDS

VARIABLE WINDS

- 1 The variable winds can be further sub divided into
 - Seasonal winds
 - Local winds

- 2 Alternatively, atmospheric motion or wind movement can be subdivided into three categories (another system of classification).

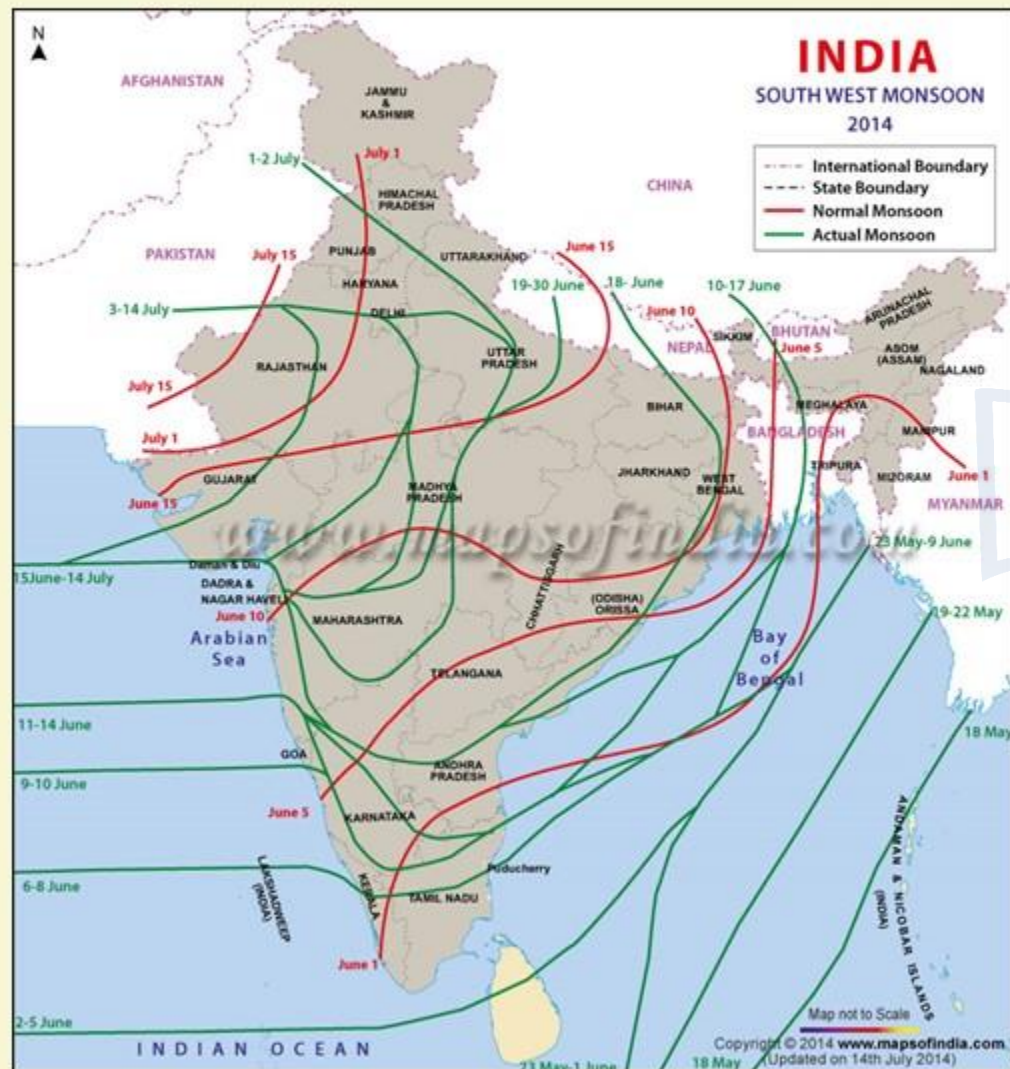
1. **Primary circulation** including planetary wind systems, which are related to global pressure belts (e.g. Trade winds, Westerlies and Polar Easterlies)
2. **Secondary circulation** consisting of cyclone, anticyclones, monsoons and air masses.
3. **Tertiary circulation** include local winds.



CLASSIFICATION OF WINDS

VARIABLE WINDS

1. SEASONAL WIND



1 The pattern of wind circulation is modified in different seasons due to the shifting of regions of maximum heating, pressure and wind belts.

2 The most pronounced effect of such a shift is noticed in the monsoon especially over South East Asia.

CLASSIFICATION OF WINDS

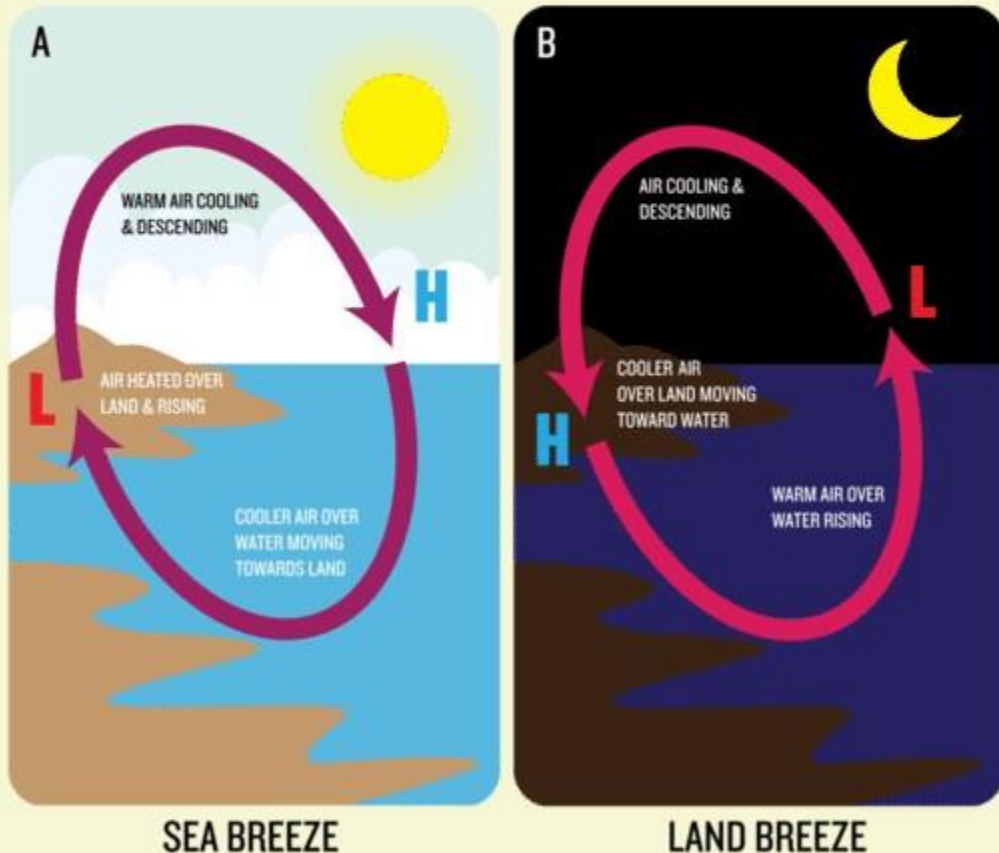
VARIABLE WINDS

2. LOCAL WIND

1 Differences in the **localised heating and cooling** of the Earth's surface and topographical differences can create several local or regional winds.

2 The most prominent among them being

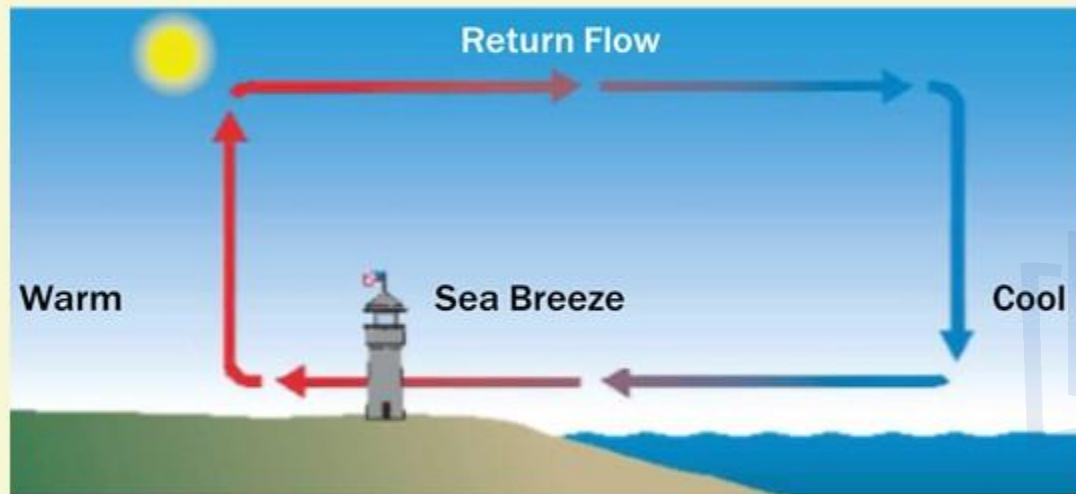
- Land and Sea Breezes
- Mountain and Valley breezes
- Chinook and Foehn
- Harmattan
- Sirocco
- Mistral
- Blizzard



CLASSIFICATION OF WINDS

LOCAL WINDS

1 LAND AND SEA BREEZES



1 The land and sea breezes representing a **complete cycle of diurnal winds** are in fact monsoon winds at a local scale because they change the direction twice in every 24 hour period.

2 These local diurnal monsoon winds are very commonly known as Land and Sea Breezes.

3 They are **found in coastal areas** where in sea breeze flows from sea to land during daytime and Land breeze moves from land to sea during night due to the differential heating and cooling of land and water.

LOCAL WINDS

1 LAND AND SEA BREEZES

SEA BREEZE

1 Because of the **high specific heat capacity of water**, it gets heated or cooled much more slowly than land.

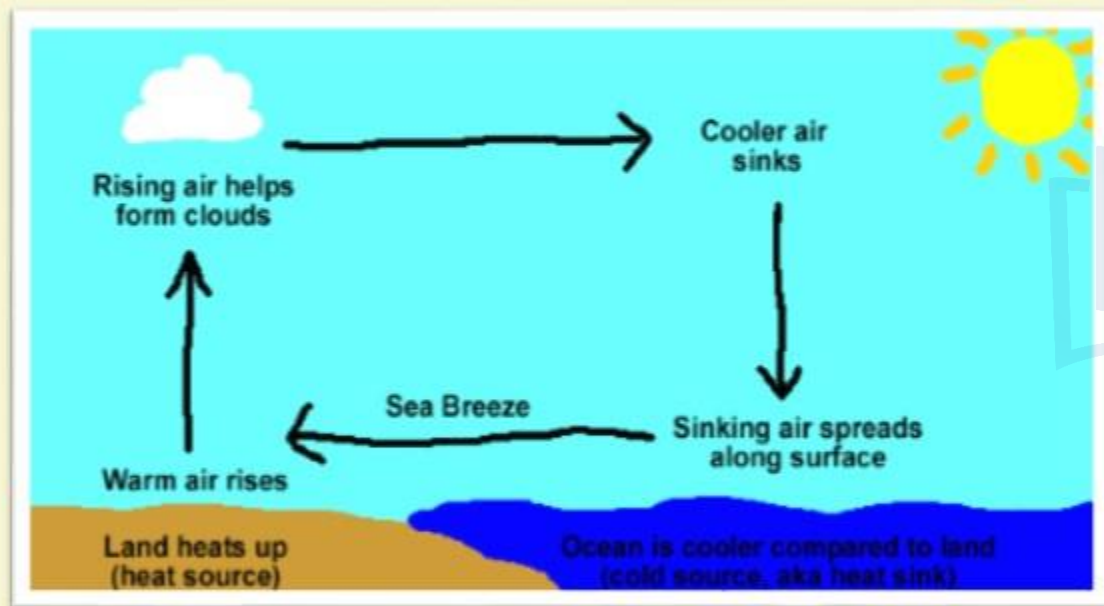
2 Land thus gets heated more quickly than the adjacent sea during the daytime. As a result, the warm air over the adjacent land expands and there is low pressure developed while high pressure is developed over the adjacent sea.

3 The **pressure gradient causes circulation** of relatively cool air from the sea to the adjacent land which is known as sea breeze.

4 Sea Breeze begin to flow usually between 10-11 AM and **become most active in early afternoon usually between 1-2 PM** and gets terminated by 8 PM at night.

LOCAL WINDS

1 LAND AND SEA BREEZES



SEA BREEZE

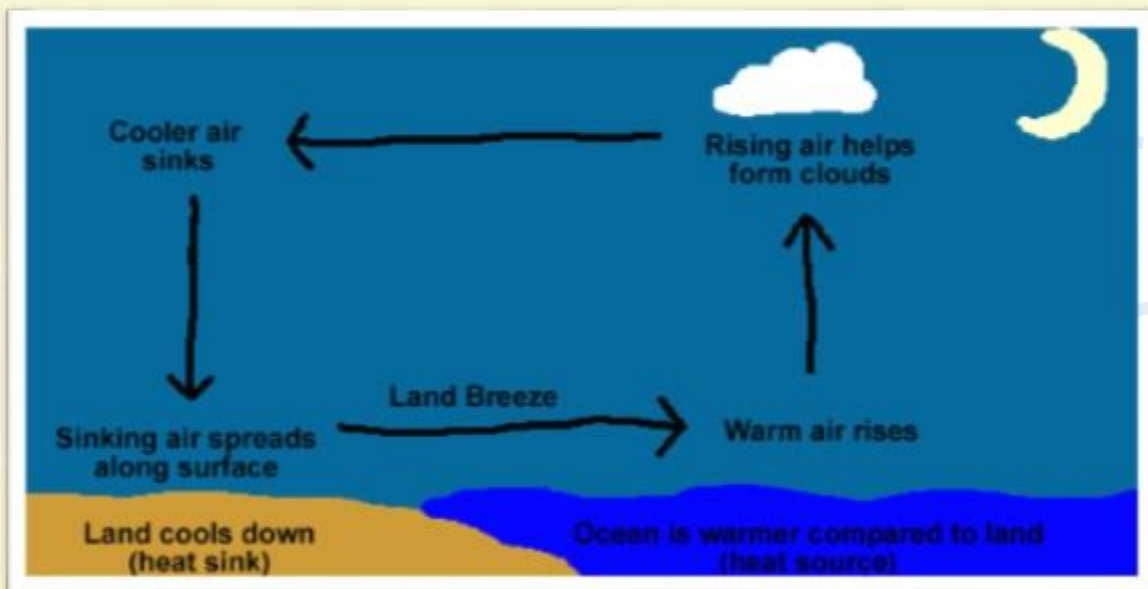
5 The cooling effect of Sea breeze reaches 50 to 65 kms inland in the tropical regions while 15 to 50 kms in the middle latitudes.

6 Sea breeze have cooling effect on coastal lands with the result weather becomes pleasant. Sea Breezes are most active during the summer season.

LOCAL WINDS

1 LAND AND SEA BREEZES

LAND BREEZE



1 After sunset the sea breezes are weakened because the day time low pressure over the land is weakened due to rapid loss of heat through outgoing longwave radiation.

2 Consequently, the position of high pressure and low pressure is reversed.

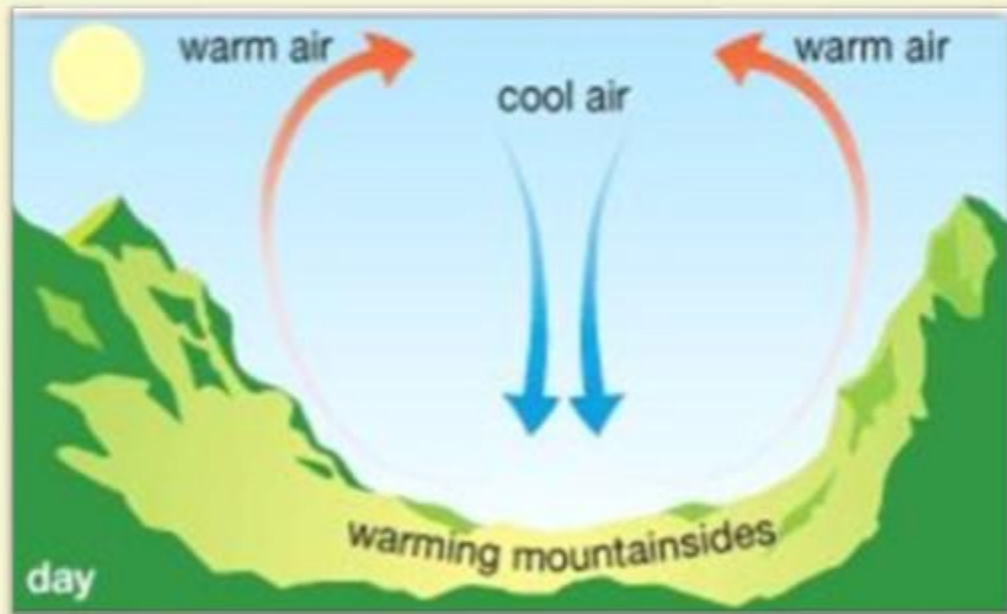
3 Now, high pressure is developed on land against the low pressure on the adjacent sea. This causes the air to move from land to sea during night, which is known as Land Breeze.

4 Land Breezes are comparatively weaker than sea breezes. These are dry winds.

CLASSIFICATION OF WINDS

LOCAL WINDS

2 MOUNTAIN AND VALLEY BREEZES

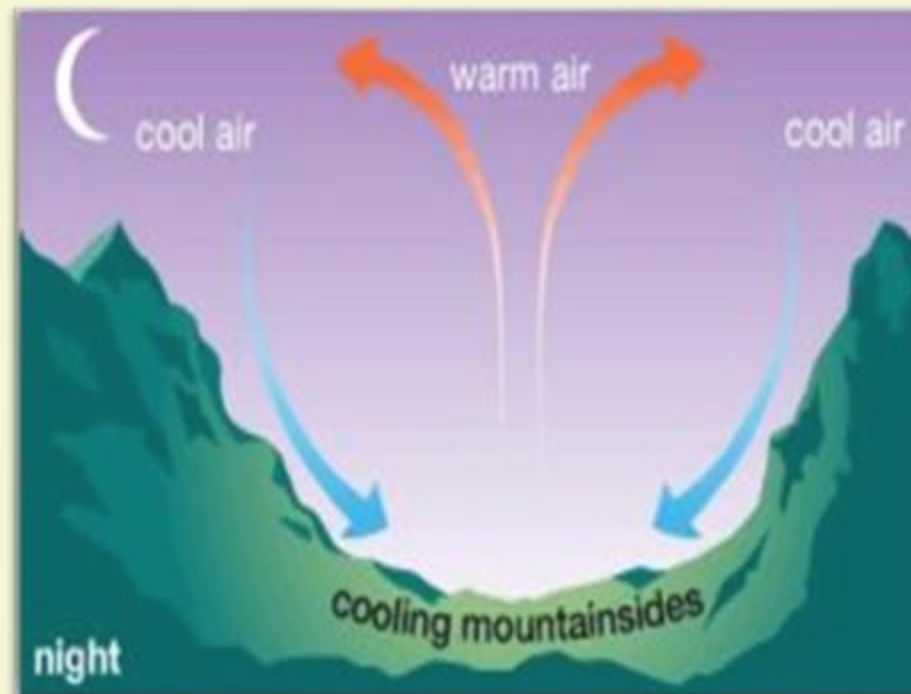


- 1 Mountain and Valley breezes are, in fact, local as well as diurnal winds.
- 2 The mountainous regions are **more heated through insolation during the daytime** than the free atmosphere at the same elevation. Consequently, the warm air moves upward.
- 3 This upslope moving breeze during daytime is called **valley breeze**.
- 4 Valley breezes reaches mountain peaks and yield precipitation through cumulus clouds.

CLASSIFICATION OF WINDS

LOCAL WINDS

2 MOUNTAIN AND VALLEY BREEZES



5 The valley slopes and upper parts are cooled due to loss of heat through outgoing longwave radiation during night-time and thus cool air descends through the valley slopes. Such winds are called mountain breeze or down valley.

6 The mountain breeze cause the **inversion** of temperature in the valleys.

7 That is why the **valley floors** are characterised by frost during night while the upper parts are free from frost in cold areas.



FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE

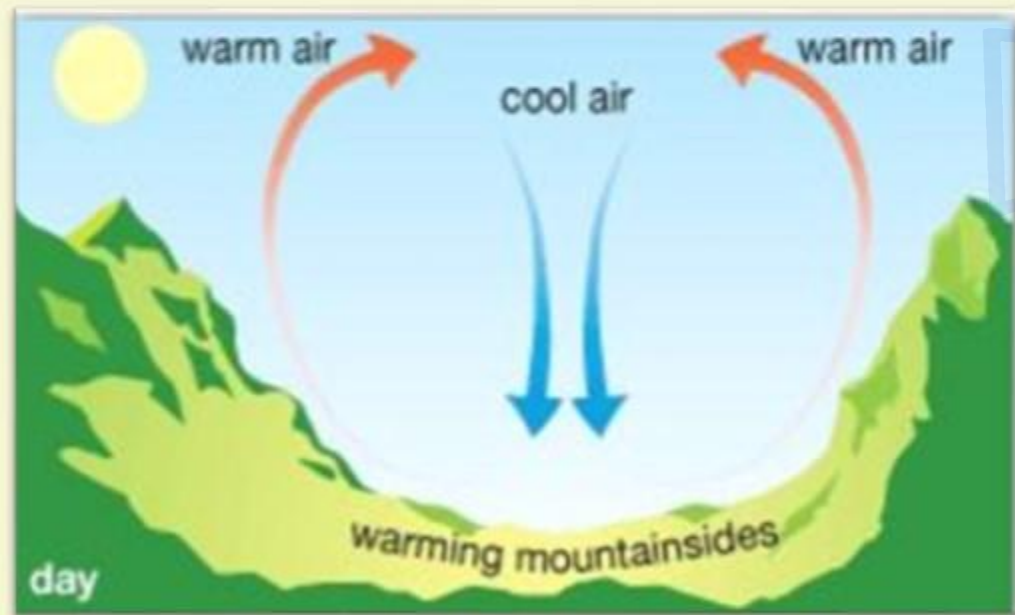


Read Module 12 for enhanced clarity

FACTOR : 6

PREVAILING WINDS

VALLEY BREEZE



- 1 In the day, the mountain top heats up faster than valley creating a thermal low at mountain top.
- 2 Hence there would be an upslope movement of air from the valley towards the hill top known as the valley breeze.
- 3 Valley breezes reach mountain peaks and yield precipitation through cumulus clouds.



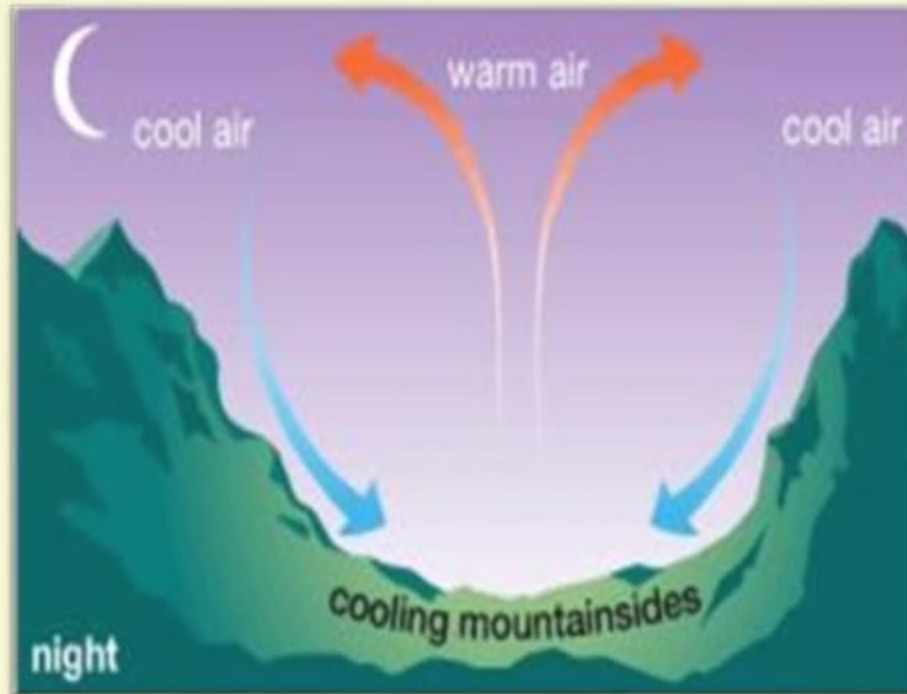
FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



Read Module 12 for enhanced clarity

FACTOR : 6

PREVAILING WINDS



MOUNTAIN BREEZE

- 1 In the night, the mountain top cools up faster creating a thermal high.
- 2 This creates a downslope movement of cool air from the hill top towards the valley known as the mountain breeze.

CLASSIFICATION OF WINDS

LOCAL WINDS

3

CHINOOK AND FOEHN

- 1 Warm and dry local winds blowing on the leeward sides of the mountains are called Chinook in the USA and Foehn in Switzerland.



- 2 The Chinook wind ascend through the western slopes of Rockies and after reaching certain height it become saturated and yield precipitation.

- 3 The winds then descend through the eastern slope of the Rockies and become warm and dry giving birth to Chinook winds.

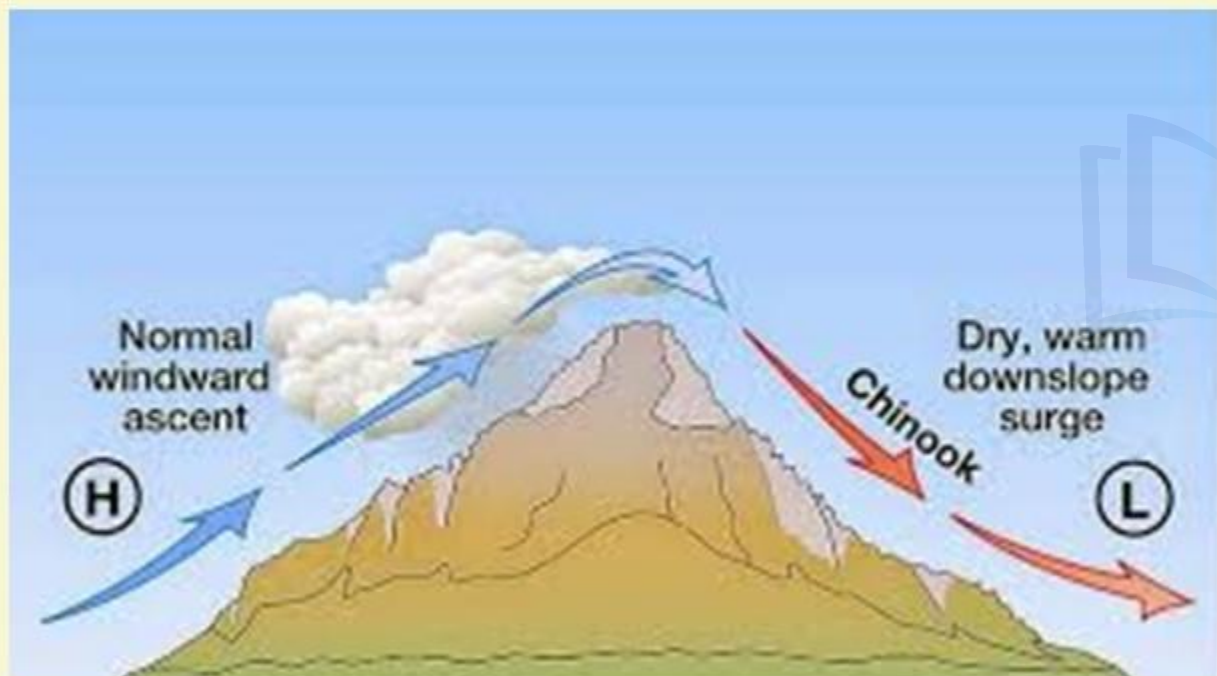
- 4 Chinook winds are more common during winter and early spring along the eastern slopes of Rocky mountains from Colorado (USA) to British Columbia (Canada).

CLASSIFICATION OF WINDS

LOCAL WINDS

3

CHINOOK AND FOEHN



4

Normally the actual temperature of the affected areas is **below freezing point** during winter months.

5

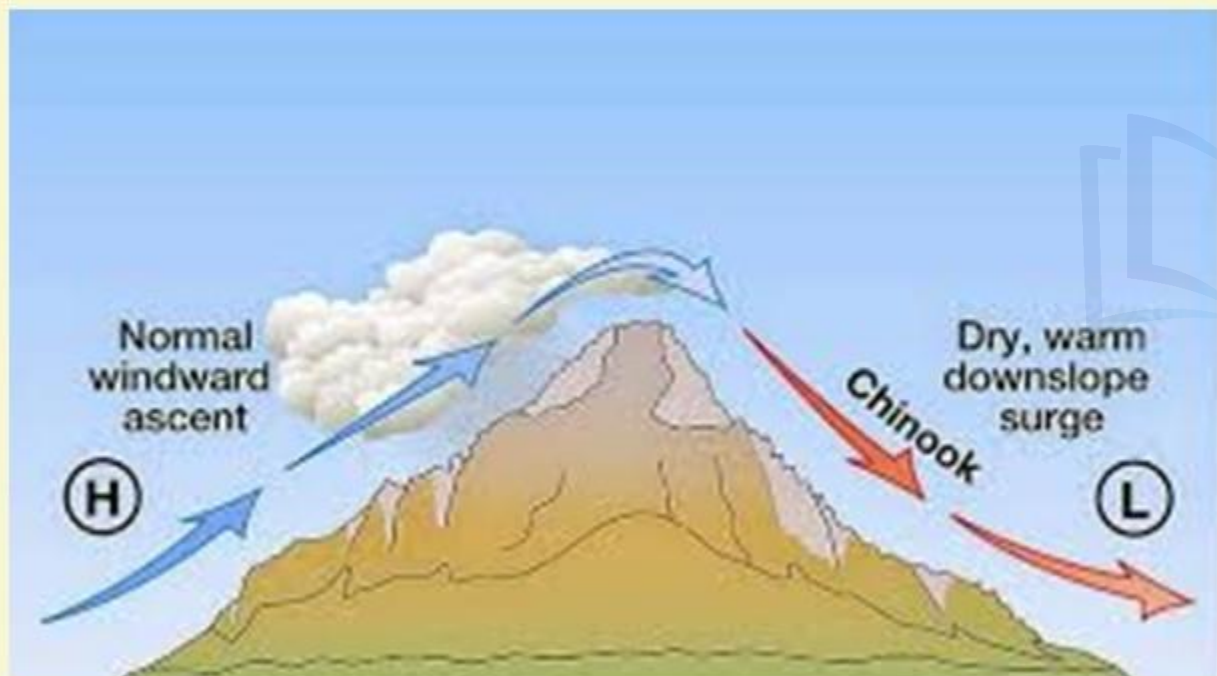
With the arrival of Chinooks the temperatures in those regions rises by **4.4⁰ C within 24 hours**. Sometimes the temperatures rises even within few minutes.

CLASSIFICATION OF WINDS

LOCAL WINDS

3

CHINOOK AND FOEHN



6 Thus, snow present on the ground surface melts away due to sudden rise in temperature as if by magic. This is why Chinook is also called as Snow Eater.

7 The impact of Chinooks is that green pastures gets opened up in a narrow strip along the eastern slopes of Rockies even during the winter season.

8 The rise in the temperature due to Chinook helps in early sowing of spring wheat in the USA.

CLASSIFICATION OF WINDS

LOCAL WINDS

3

CHINOOK AND FOEHN

- 9** A warm and dry wind similar to Chinook is called Foehn along the slopes of the Alps mountains in Switzerland.
- 10** The weather becomes quite pleasant in the valleys due to melting point of snow due to increase in temperature by 40° F after the arrival of Foehn winds.
- 11** This is why valleys of Switzerland are called “Climatic Oasis” during winter season.
- 12** These winds helps in early sowing of wheat, ripening of grapes and check autumn frost.

CLASSIFICATION OF WINDS

LOCAL WINDS

PROBABLE QUESTION FOR MAINS

Q 3 WHAT ARE LOCAL WINDS? HOW DO THEY AFFECT THE LOCAL WEATHER CONDITIONS AND HUMAN HABITATS? **(10 Marks) (150 words)**

HINTS:

To be discussed in the next module

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